# Trading Volume Alpha


| Ruslan Goyenko | Bryan Kelly | Tobias Moskowitz | Yinan Su | **Chao Zhang** |
|---|---|---|---|---|
| *McGill* | *Yale* | *Yale* | *JHU* | HKUST (GZ) |


ML for Finance
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# Motivation

AI/ML for investments

- powerful tools for portfolio optimization

  superiority in extracting trading signals that forecast future returns

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- from prediction to implementation
  - Trading costs
  - a wedge between academic and practitioner research
- from statistical loss to economic loss

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Key message

- beyond return forecasting
- economic value

# Introduction

Portfolio optimization chiefly focuses on return and risk prediction

In implementation: trading costs

- with trading costs, a key trade-off:
  passive vs. aggressive trading, depending on trading costs forecasts

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- with trading costs, a key trade-off:
  passive vs. aggressive trading, depending on trading costs forecasts
- price impact depends on how much to trade, how much others are trading ...

This paper: focus on trading volume

forecast volume $\rightarrow$ portfolio strategy $\rightarrow$ net-of-cost "alpha"

# Why trading volume?

- individual trade sizes are hard to obtain

- market liquidity proxied by volume, unknown ex ante
- a general yet critical component of trading costs modeling

- Kyle (1985); Frazzini, Israel, and Moskowitz (2018):

$$PriceImpact_{a,i,t} \propto ParticipationRate_{a,i,t} := \frac{\$Traded_{a,i,t}}{\$Volume_{i,t}}$$

  volume is generic across investors ($a$), exogenous, observable

- volume forecast informs trading decision
  volume $\uparrow \Rightarrow$ price impact $\downarrow \Rightarrow$ trade more aggressively (vice versa)

# Introduction

Main contributions:

- forecast volume $\rightarrow$ portfolio strategy $\rightarrow$ net-of-cost "alpha"

- new dataset, neural network methods, and new benchmarks
- transfer statistical learning into finance problem solving
- substantial economic gain, comparable to return prediction

→ Volume prediction from statistical perspective

- Portfolio implementation based on volume prediction
- Investment performance evaluation

# Daily stock-level dollar trading volume

Data structure

- unit of observation: stock, day ($i, t$ panel)
- sample period from 2018 to 2022, 5 years or 1,200+ days
  3 year train + 2 year testing
  4,700+ stocks in total, avg 3,500 observed stocks per day,
  4,400,000+ observations in total

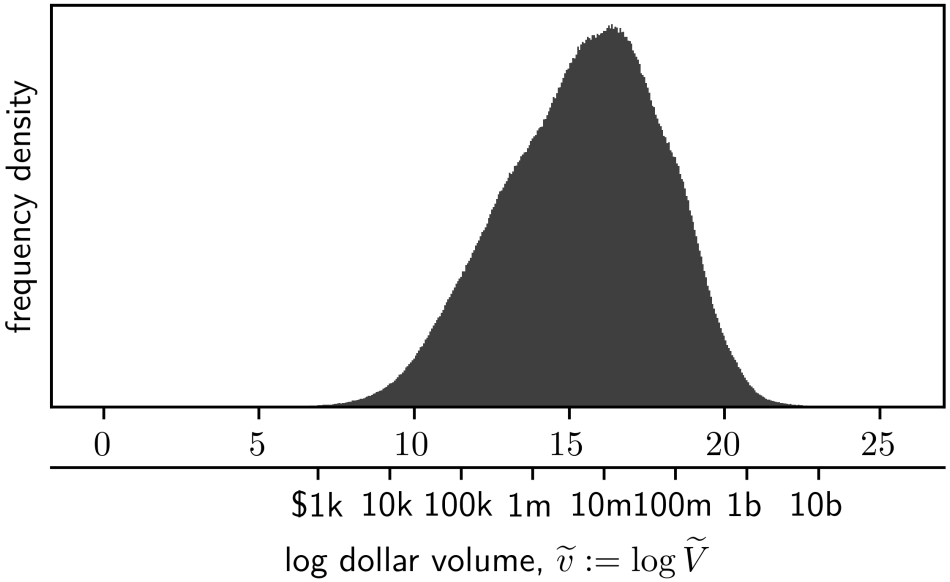
# Predictors

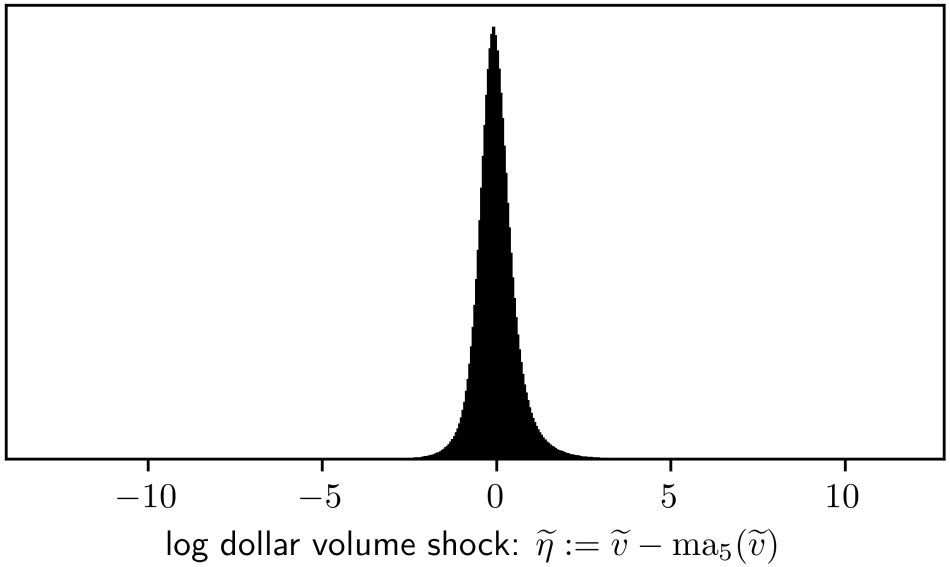
Predictors, $X_{i,t}$, 175 in total, 4 categories:

- technical signals:
  return and volume lagged moving averages
- firm characteristics:
  size, b/m, ..., developed for return prediction (JKP dataset)
- deterministic calendar events:
  triple witching[1], index rebalancing, early closing etc.
- scheduled earnings announcements

[1] Triple witching is the quarterly expiration of stock options, stock index futures, and stock index options contracts all on the same day.

# Dollar volume, log dollar volume, and its shock




- dollar volume $\widetilde{V}$; log dollar volume $\widetilde{v}$
- take $\mathrm{ma}_5$ as baseline throughout the paper
  predict log dollar volume shock $\widetilde{\eta} := \widetilde{v} - \mathrm{ma}_5$
  - like predicting return := log price - log yesterday's price.

# Neural network tools

Run (ols and nn) regressions of $\widetilde{\eta}_{i,t}$ on $X_{i,t}$

Transparent and standard machine learning implementation

- nn (neural network):
  3-layer, fully-connected, 32-16-8 ReLU nodes
- rnn (recurrent neural network):
  to capture time series predictive dependency,

$$(\widehat{\eta}_{i,t}, state_{i,t}) = rnn(X_{i,t}, state_{i,t-1})$$

  architecture: lstm as the first hidden layer, all else the same

# Prediction results, OOS $R^2$ (%)

| cumulative | tech | fund-1 | fund-2 | calendar | earnings |
|---|---|---|---|---|---|
| # of predictors | 8 | 14 | 161 | 165 | 175 |
| A: $R^2$ relative to $\widetilde{\eta}$ $(\widetilde{v} - \mathrm{ma}_5)$ | | | | | |
| $\mathrm{ma}_5$ | 0 | | | | |
| ols | 12.09 | 12.26 | 12.27 | 14.85 | 15.99 |
| nn | 14.31 | 14.90 | 14.42 | 17.13 | 18.45 |
| rnn | 15.80 | 16.25 | 15.47 | 18.12 | 19.86 |
| B: number of parameters | | | | | |
| $\mathrm{ma}_5$ | 0 | | | | |
| ols | 9 | 15 | 162 | 166 | 176 |
| nn | 961 | 1,153 | 5,857 | 5,985 | 6,305 |
| rnn | 6,049 | 6,817 | 25,633 | 26,145 | 27,425 |

- Volume prediction from statistical perspective
→ Portfolio strategy based on volume prediction
- Investment performance evaluation

# Portfolio optimization problem

Trade-off: track a target portfolio vs. avoid trading cost:

Given starting \$ position $x^0$, target \$ position $x^*$, choose new position $x$ or trade amount $\Delta x := x - x^0$ (for each stock each day)

$$\min_{\{x_{i,t}\}} \sum_{i,t} \left( TrackError_{i,t} + TradeCost_{i,t} \right), \quad \text{s.t. } x_{i,t} \in t-1 \text{ info}$$

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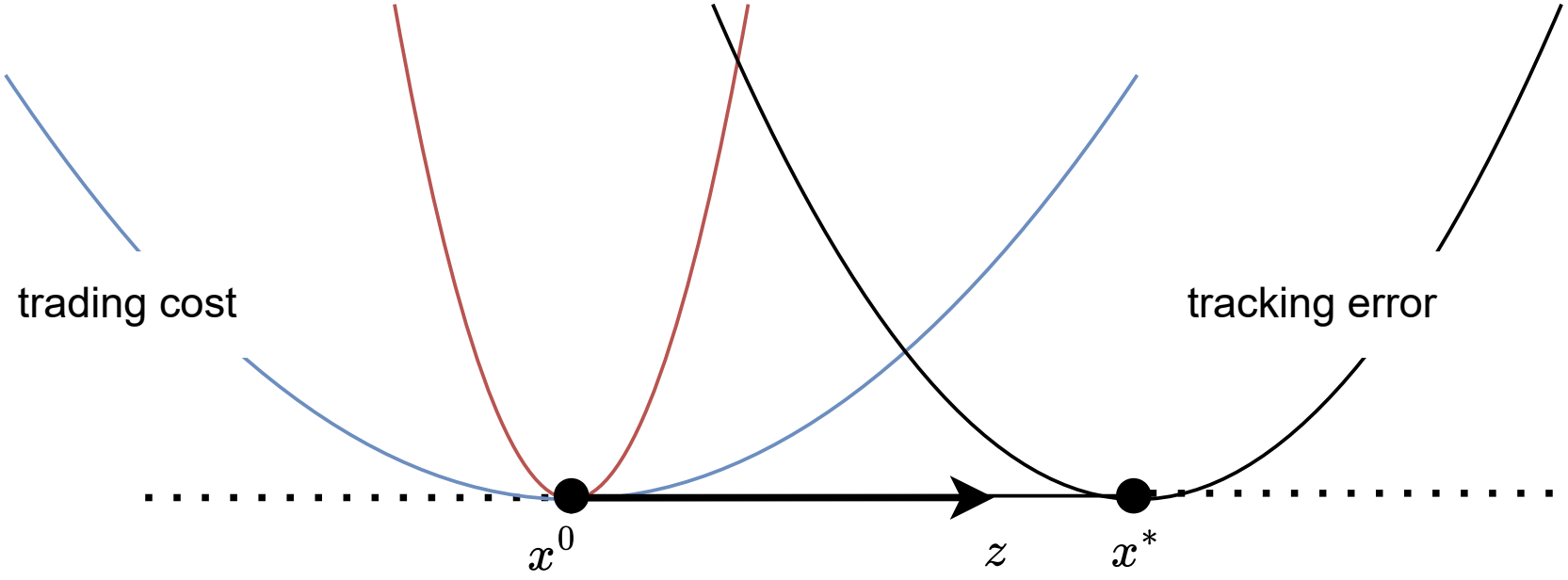


- choose $z \in [0, 1]$ (stay put vs. trade all the way)
- predicting volume well benefits this optimization

  intuition: trade more aggressively if believe $\widetilde{v}$ high, vice versa

# Tracking error modeling

$$TrackError := \frac{1}{2}\mu(x^* - x)^2$$

- Microfoundation: from a mean-variance optimization problem,
  ▸ microfoundation
  $x^*$ embeds return signals, AUM, leverage etc.
- $\mu$ controls the penalty for being away from the target
  like a hyperparameter
  larger $\mu \Rightarrow$ trade more aggressively in general

# Trading cost modeling

$$TradeCost := \frac{1}{2}\widetilde{\lambda}(x - x^0)^2$$
$$\widetilde{\lambda} := 0.2/\widetilde{V}$$

Microfoundation: $PriceImpact$ linear in participation rate:

$$PriceImpact = 0.1\frac{x - x^0}{\widetilde{V}}, \quad TradeCost = PriceImpact \cdot (x - x^0)$$

E.g., buying 10% of daily volume incurs 1% price impact (Frazzini, Israel, and Moskowitz, 2018)

# Combined objective

Minimize the expectation of $TradeCost + TrackError$ conditional on information

$$\min_x \mathbb{E}\left[\frac{1}{2}\widetilde{\lambda}(x - x^0)^2 + \frac{1}{2}\mu(x^* - x)^2 \middle| \text{conditioning information}\right]$$

Importantly, $TradeCost$ is unknown ex ante when choosing $x$: $\widetilde{\lambda}$ depends on $\widetilde{V}$.

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Importantly, $TradeCost$ is unknown ex ante when choosing $x$: $\widetilde{\lambda}$ depends on $\widetilde{V}$.

Key point: predicting $\widetilde{V}$ well helps solving this problem.
Intuition: more adaptive to liquidity condition:
trade more aggressively if believe $\widetilde{V}$ high, patiently otherwise

Next: analyze and solve the problem to demonstrate this point and show the economic gains

# Objective function analysis

Isolate target trade size:

Define trading rate $z := \frac{x-x^0}{x^*-x^0}$, then the target becomes

$$\begin{aligned}&\frac{1}{2}\widetilde{\lambda}(x-x^0)^2+\frac{1}{2}\mu(x^*-x)^2\\=&\frac{1}{2}\left(x^*-x^0\right)^2\big(\underbrace{\widetilde{\lambda}z^2+\mu(1-z)^2}_{loss^{\text{econ}}(\widetilde{v},z;\mu)}\big)\end{aligned}$$

New problem: minimize expected economic loss by choosing $z$

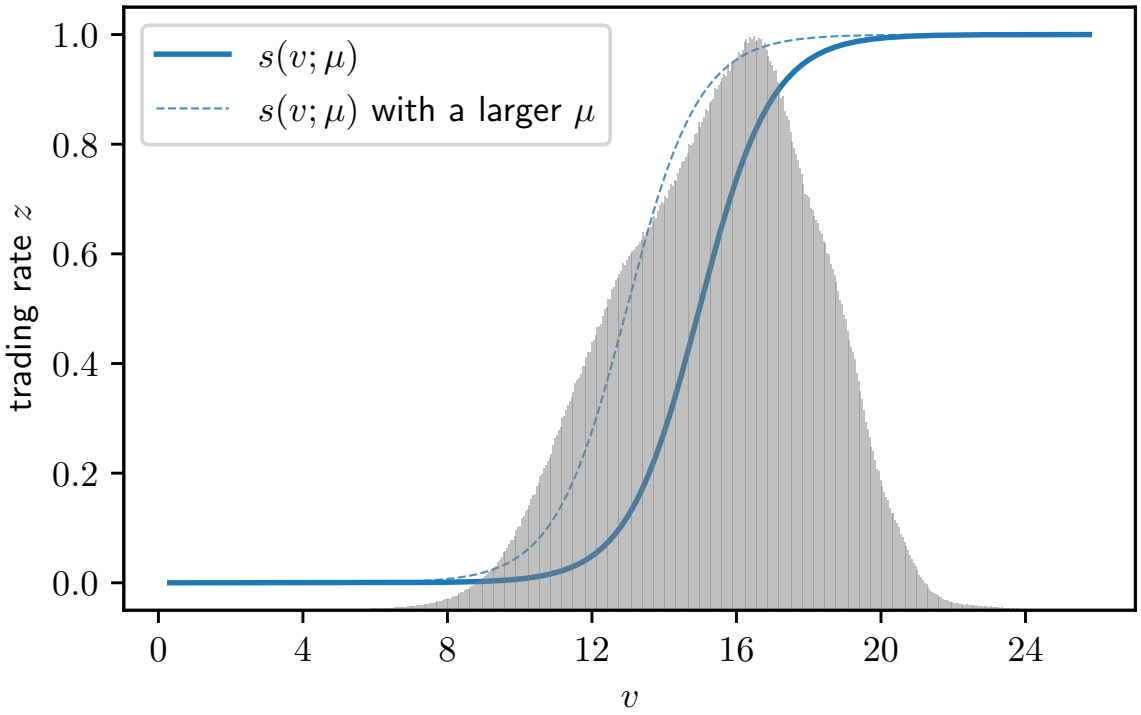
$$\min_z \mathbb{E}\left[loss^{\text{econ}}(\widetilde{v},z;\mu)|X\right]$$

- same problem regardless of tracking target (will experiment with different targets)
- for now, choose trade rate $z \in [0,1]$, solve the normalized problem

# $loss^{\text{econ}}$ analysis: $z$ and $v$ relationship

say we believe $\widetilde{v} = v$, that is $\lambda = 0.2/\exp(v)$:

$$\arg\min_z loss^{\text{econ}} = \frac{\mu}{\mu + \lambda} = \text{sigmoid}\big(v - \log(0.2/\mu)\big) \to s(v;\mu)$$




- a forecast of $v$ corresponds to a choice $z$,
  conversely, a choice $z$ implies an (underlying) forecast of $v$

# Approach 1, statistical learning
# plug-in stats forecast $\widehat{v}$, trade $\widehat{z} = s(\widehat{v}; \mu)$

1. ML based on a statistical loss, MSE:

$$\min_{v(\cdot)} \sum_{i,t \in \text{train}} loss^{\text{least squares}}\big(\widetilde{v}_{i,t}, v(X_{i,t})\big)$$

2. plug $\widehat{v}_{i,t} = v(X_{i,t})$ into the optimal policy $s(\,\cdot\,; \mu)$
   choose trading rate $\widehat{z}_{i,t} = s(\widehat{v}_{i,t}; \mu)$

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- Evaluated by economic loss

$$\frac{1}{|\text{test}|} \sum_{i,t \in \text{test}} loss^{\text{econ}}\Big(\widetilde{v}_{i,t}, \widehat{z}_{i,t}; \mu\Big)$$

[ $loss^{\text{least squares}}(\widetilde{v}, v) = (\widetilde{v} - v)^2$, $loss^{\text{econ}} =$ ... as defined ]

# Approach 2, economic learning: directly learn $z$ to optimize economic loss

$$\min_{z(\cdot)} \sum_{i,t \in \text{train}} loss^{\text{econ}}(\widetilde{v}_{i,t}, z(X_{i,t}))$$

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Equivalently, this is also predicting volume,
because choosing $z$ implies a forecasted value $v$

$$\min_{v(\cdot)} \sum_{i,t \in \text{train}} loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_{i,t}, v(X_{i,t}))$$

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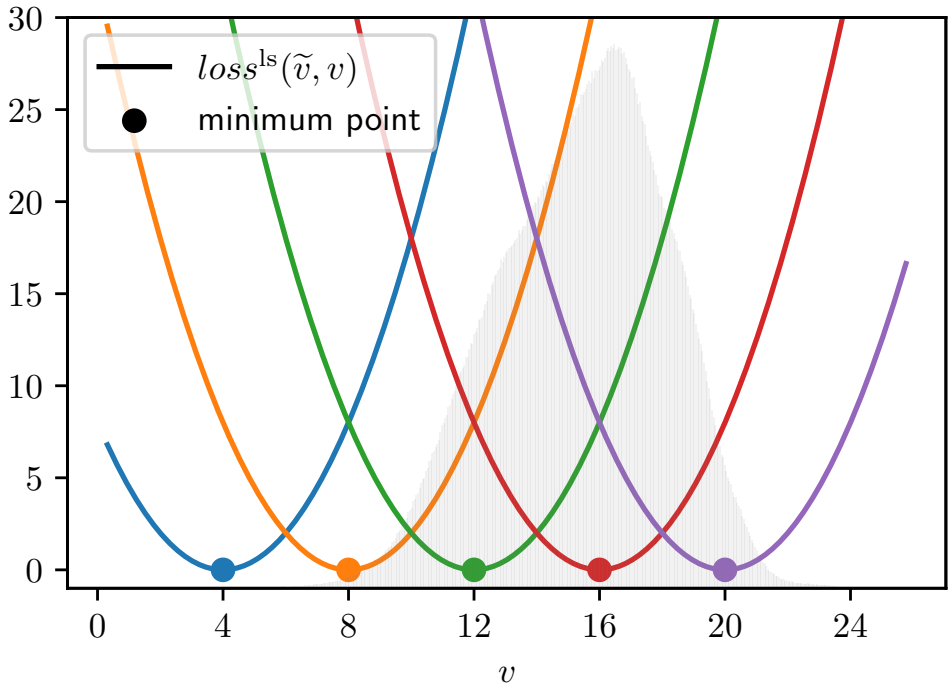
$$\min_{v(\cdot)} \sum_{i,t \in \text{train}} loss^{\text{econ}}_{\text{vv}}(\widetilde{v}_{i,t}, v(X_{i,t}))$$

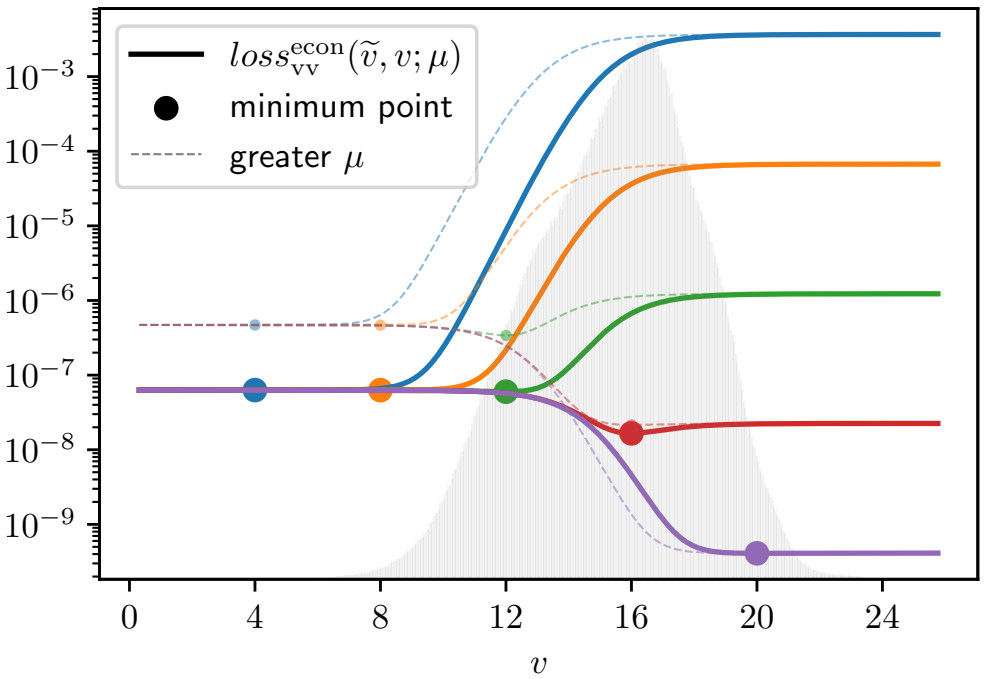
Comparing the two methods:

- Only difference is at the loss functions

  Next, loss functions comparison and implications on learning

# Comparing statistical and economic loss



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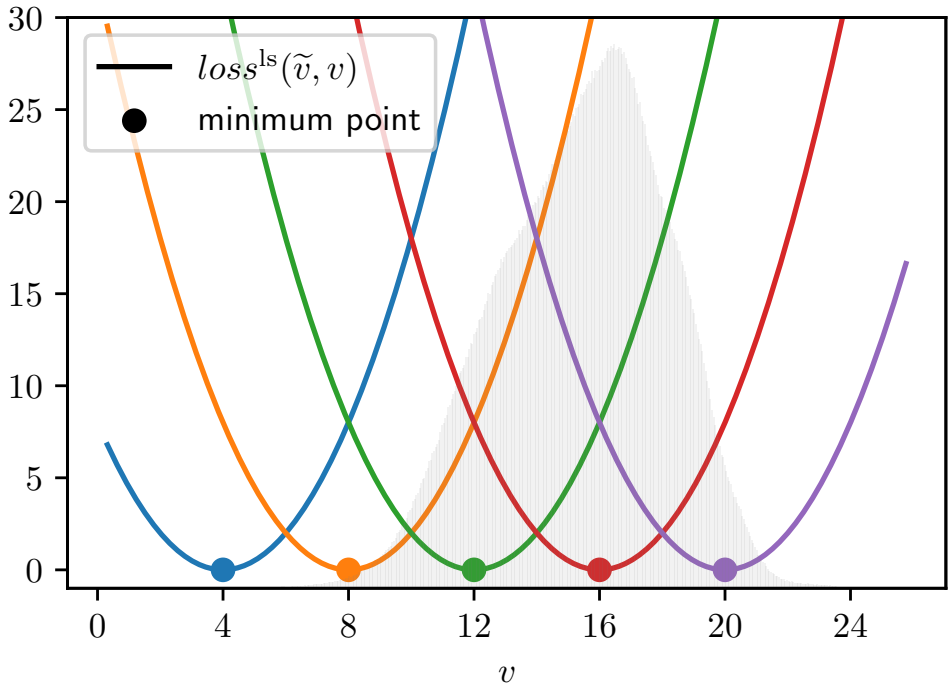
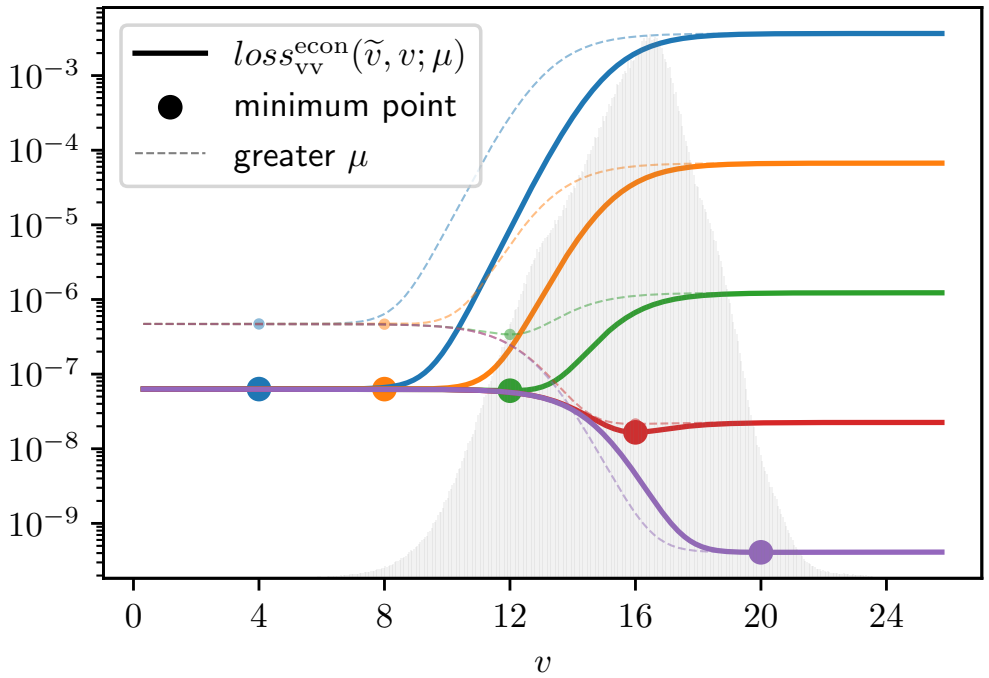



Economic loss is asymmetrical

- huge cost if actual $\widetilde{v}$ is low but predicted as high
  trade a lot when liquidity is low
- in contrast, opportunity cost for not trading is bounded

Economic objective favors conservatism to avoid overestimating volume

# Theoretical analysis

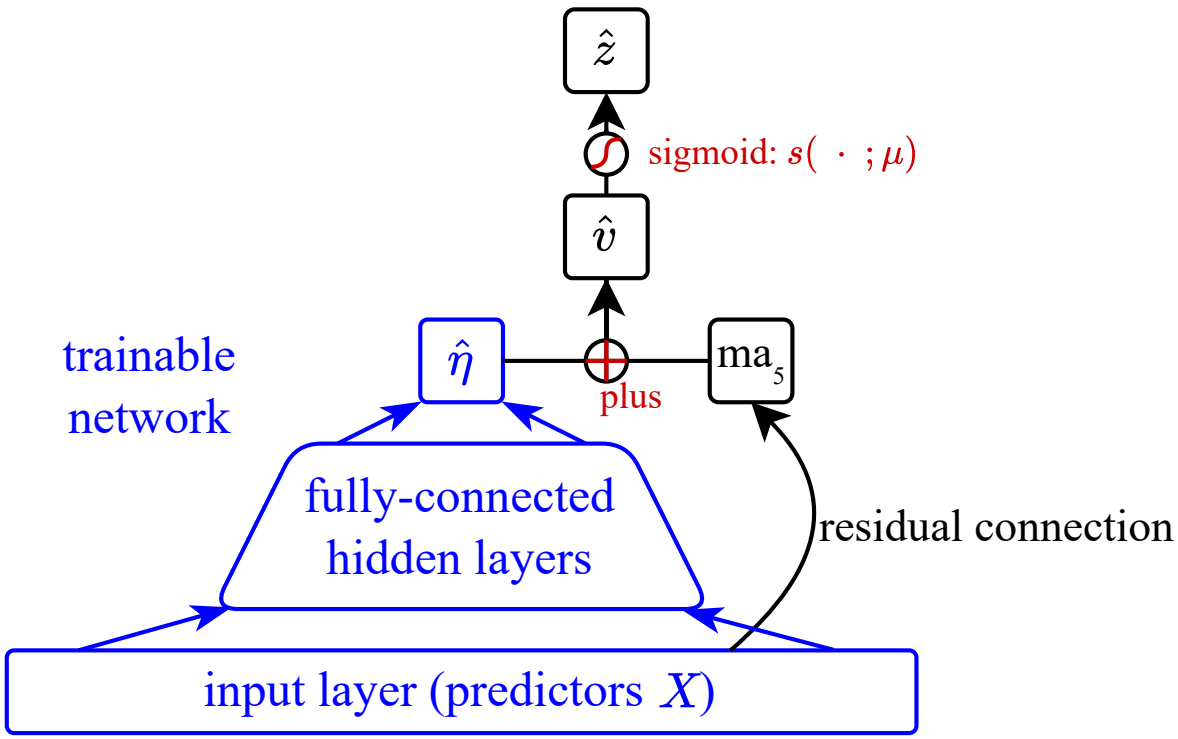
Proposition: least squares optimizer does not maximize economic objective, even in population

- well-known: conditional expectation is the least squares minimizer

$$\mathbb{E}[\widetilde{v}|X] = \min_{v \in \sigma(X)} \mathbb{E}[loss^{\text{least squares}}(\widetilde{v}, v)]$$

- how about other loss functions?
  iff the loss function is in the Bregman class (Banerjee, Guo, and Wang, 2005; Patton, 2020)
  however, $loss_{\text{vv}}^{\text{econ}}$ is not in Bregman class

# Network architecture, transfer learning



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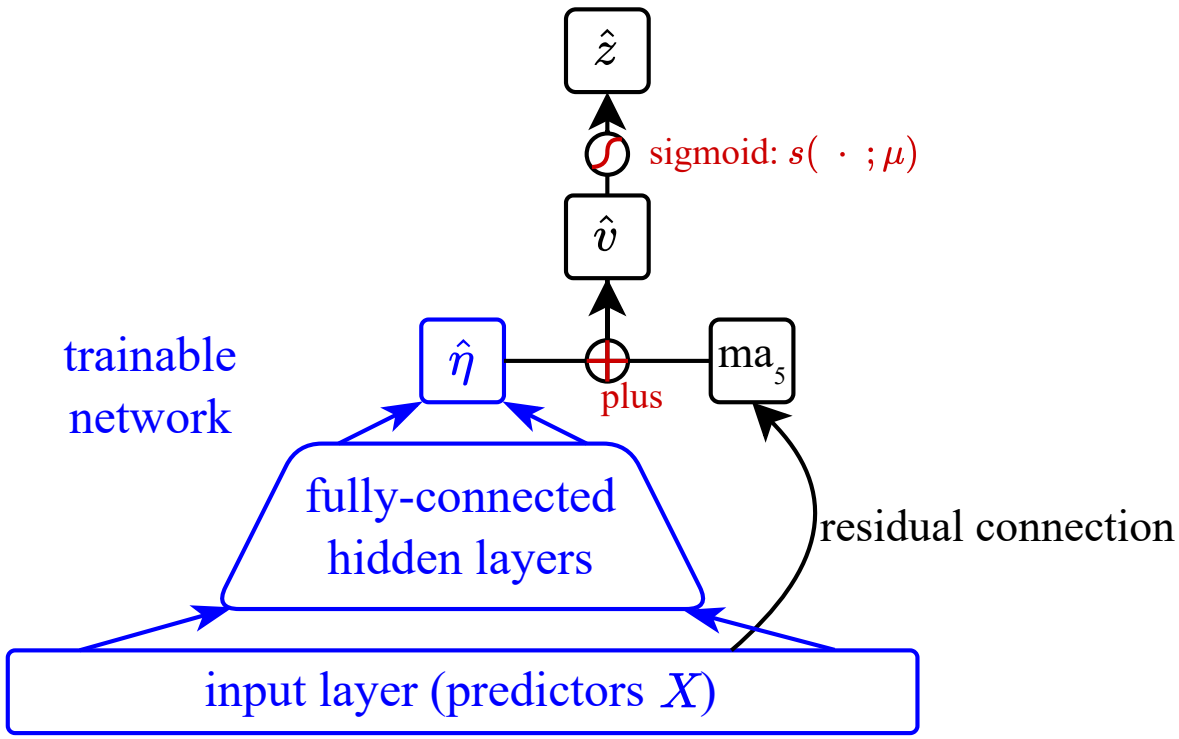


Transfer learning:

- statistical learning fits an nn model, applicable to all $\mu$ values
  serve as pre-trained foundation model
- economic learning fine-tunes the nn for each $\mu$
  fine-tuning for downstream tasks, since $loss^{\mathrm{econ}}$ is indexed by $\mu$

# Transfer learning, summary

- Method 1, statistical learning:
  - run ML regressions of $\widetilde{v}$ onto $X$, learn $v^{\mathrm{nn}}(X)$ in training sample

$$\min_{\theta} \sum_{i,t \in \mathrm{train}} (\widetilde{v}_{i,t} - v^{\mathrm{nn}}(X_{i,t};\theta))^2$$

  - plug-in and trade $s(v^{\mathrm{nn}}(X_{i,t};\theta^*),\mu)$ in testing sample
- Method 2, economic learning:
  - directly learn $z$:

$$\min_{\theta} \sum_{i,t \in \mathrm{train}} loss^{\mathsf{econ}}(\widetilde{v}_{i,t}, z^{\mathrm{nn}}(X_{i,t};\theta);\mu)$$

  - trade $z^{\mathrm{nn}}(X_{i,t};\theta^*_{\mu})$ in testing sample

Method 1 fits 1 nn model, applicable to all $\mu$ values
Method 2 fine-tunes the nn for each $\mu$, because $loss^{\mathrm{econ}}$ is indexed by $\mu$

- Volume prediction from statistical perspective
- Portfolio strategy based on volume prediction

→ Investment performance evaluation

# Trading experiments, setup

Apply the strategies ($\widehat{z}$) to track various sets of trading targets $\left\{x^*_{i,t}\right\}$ that mimic different trading tasks, including

1. simulated quantitative trading strategy
   implementing **before-cost** high-Sharpe strategies
2. factor zoo portfolios (JKP factors)
   implementing different investment styles

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Experiment procedure: recursively track the supplied target

- given $x^0_{i,t}, x^*_{i,t}$, trade $x_{i,t} = x^0_{i,t} + \widehat{z}_{i,t}(x^*_{i,t} - x^0_{i,t})$
- roll to $x^0_{i,t+1} := x_{i,t}(1 + r^{\mathrm{f}}_t + r_{i,t})$
- trade $x_{i,t+1}$ to track $x^*_{i,t+1}$, and so on ...

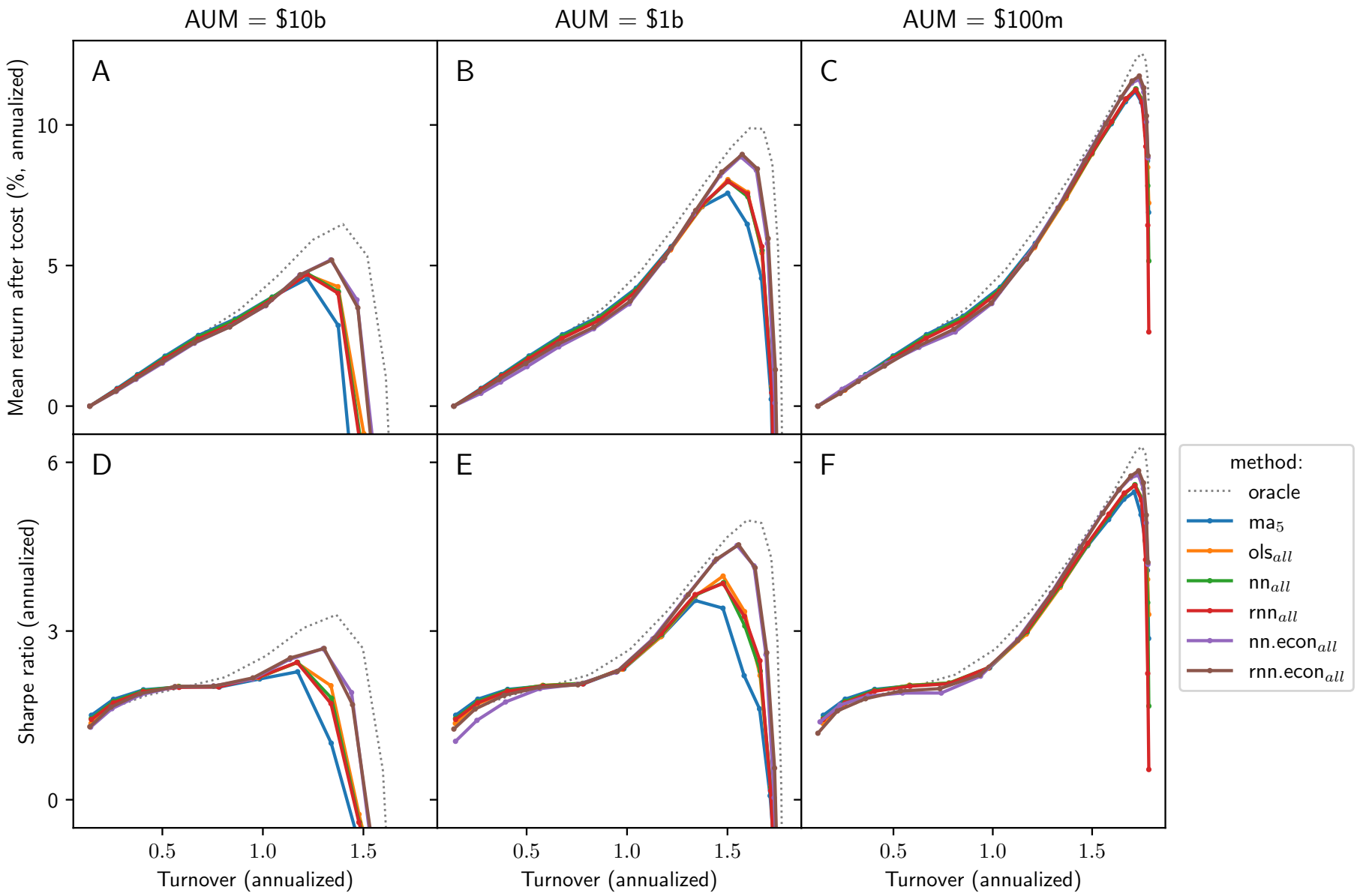
Evaluate the investment outcome of $\{x_{i,t}\}$ in the testing sample

# Experiment 1, quantitative trading task

Construct targets $\left\{x^*_{i,t}\right\}$ with unrealistically high (before-cost) investment performance

- Sharpe ratio $\approx 7$
- blow up to various AUM magnitudes

# Performance across $\mu$, all 175 predictors



# Performance with $\mu$ hyper-parameter tuning

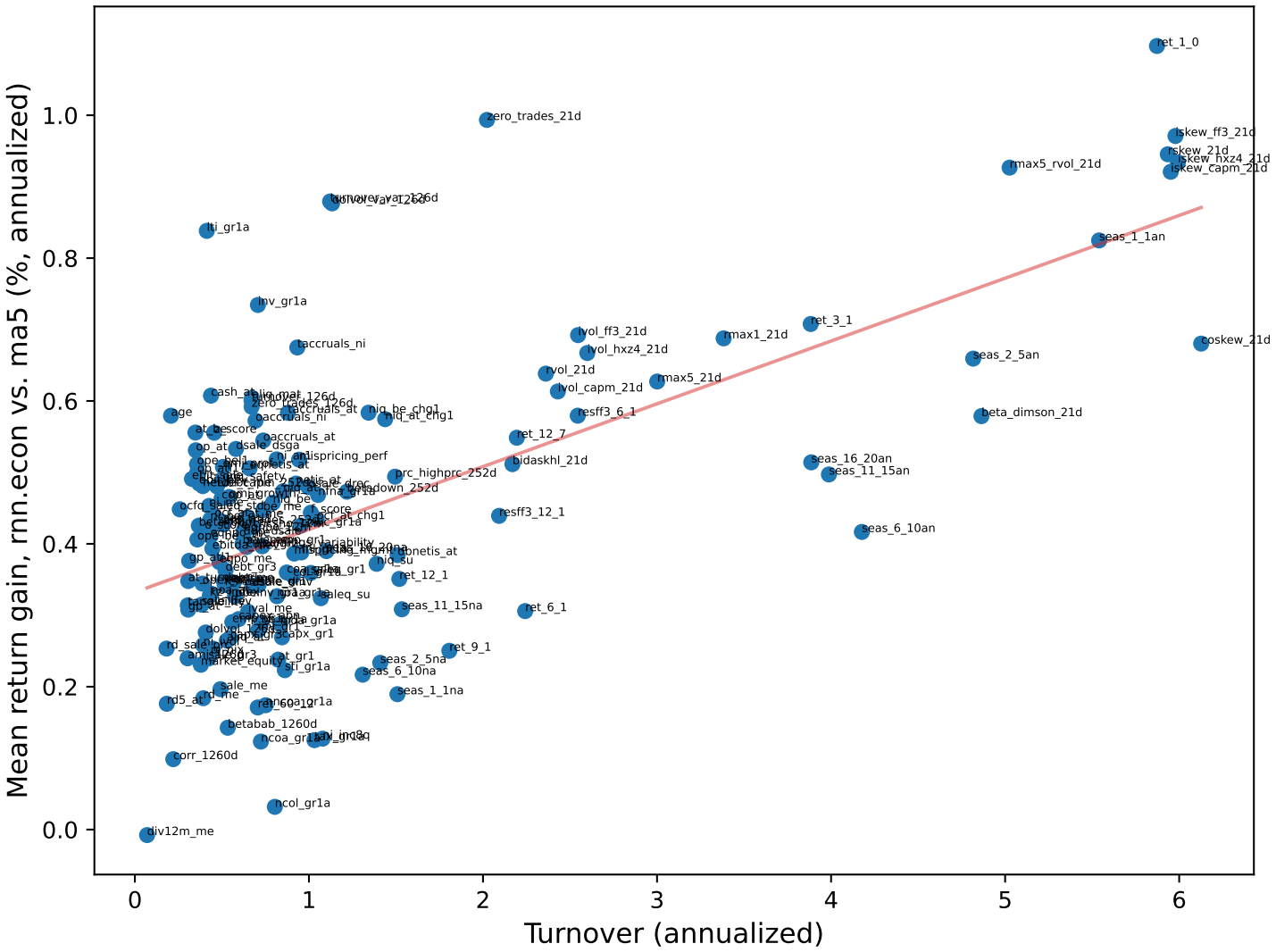
| A. Mean return (%, annualized) | | | | | B. Sharpe ratio (annualized) | | | |
|---|---|---|---|---|---|---|---|---|
| AUM | \$10b | \$1b | \$100m | \$10m | \$10b | \$1b | \$100m | \$10m |
| $\text{ma}_5$ | 3.88 | 6.47 | 11.19 | 13.20 | 2.00 | 2.21 | 5.47 | 6.55 |
| $\text{ols}_{tech}$ | 3.82 | 7.60 | 11.28 | 13.14 | 2.16 | 3.32 | 5.59 | 6.59 |
| $\text{nn}_{tech}$ | 3.76 | 7.30 | 11.32 | 13.13 | 2.14 | 2.79 | 5.63 | 6.59 |
| $\text{rnn}_{tech}$ | 3.74 | 7.84 | 11.33 | 13.13 | 2.18 | 3.58 | 5.64 | 6.59 |
| $\text{nn.econ}_{tech}$ | 4.60 | 7.20 | 11.29 | 13.22 | 2.13 | 2.57 | 5.59 | 6.63 |
| $\text{rnn.econ}_{tech}$ | 4.67 | 8.69 | 11.59 | 13.30 | 2.50 | 4.32 | 5.73 | 6.66 |
| $\text{ols}_{all}$ | 3.82 | 7.60 | 11.28 | 13.14 | 2.17 | 3.35 | 5.60 | 6.59 |
| $\text{nn}_{all}$ | 3.86 | 7.44 | 11.28 | 13.13 | 2.19 | 3.09 | 5.60 | 6.58 |
| $\text{rnn}_{all}$ | 3.79 | 7.55 | 11.25 | 13.09 | 2.18 | 3.26 | 5.59 | 6.56 |
| $\text{nn.econ}_{all}$ | 4.64 | 8.87 | 11.61 | 13.29 | 2.18 | 4.24 | 5.74 | 6.66 |
| $\text{rnn.econ}_{all}$ | 4.68 | 8.95 | 11.77 | 13.30 | 2.50 | 4.53 | 5.85 | 6.68 |
| oracle | 6.47 | 9.89 | 12.54 | 13.56 | 3.05 | 4.97 | 6.28 | 6.80 |

# Experiment 2, implementing factor zoo portfolios

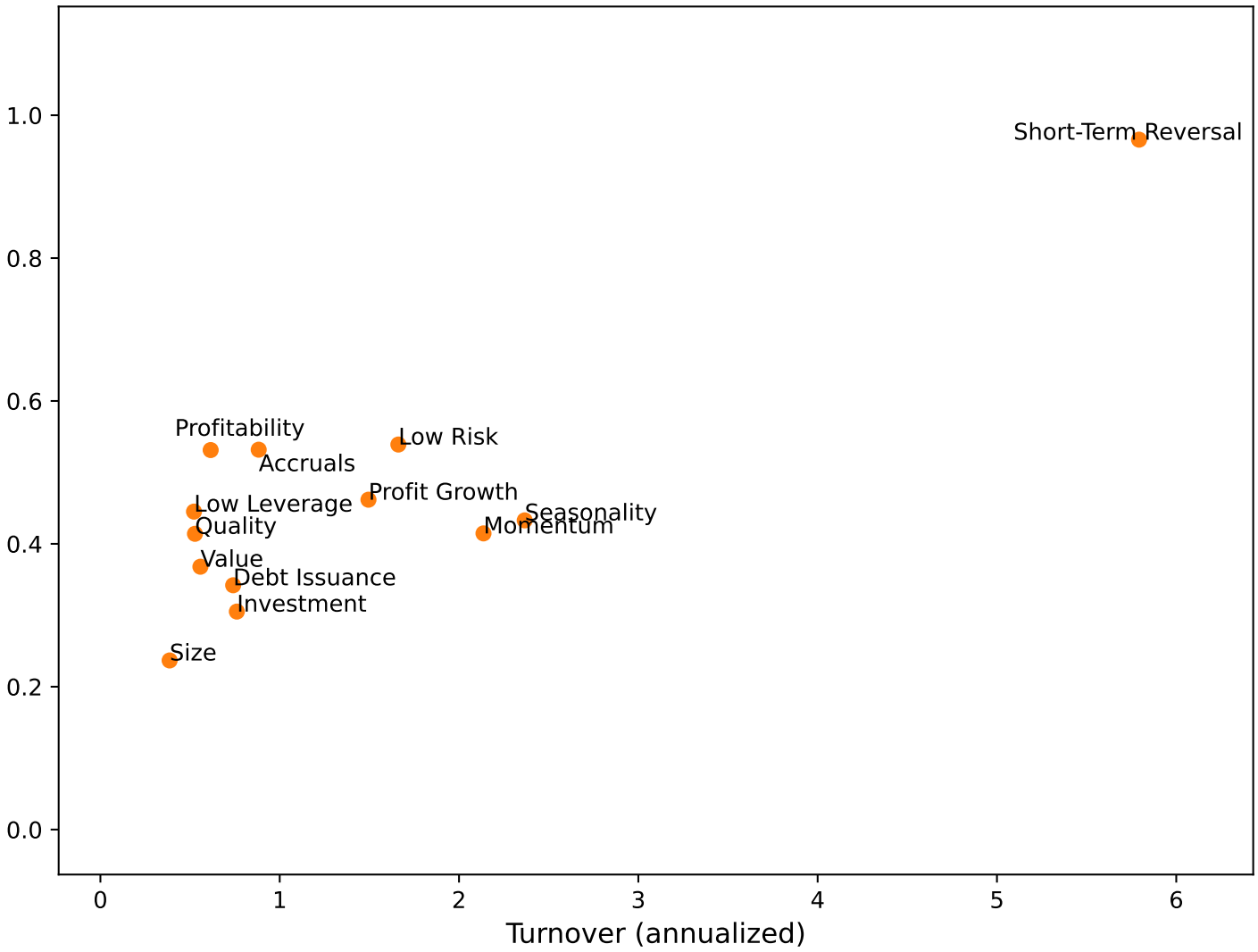
Long-short sorted portfolios for each 153 JKP characteristics

- AUM $10b, leverage 100%

# Mean return improvements in implementing factor portfolio



# Mean return improvements averaged by theme clusters



# Trading volume alpha, summary

- daily firm-level trading volume is highly predictable, especially with machine learning
- forecast volume $\rightarrow$ portfolio strategy $\rightarrow$ net-of-cost "alpha"
  economic value comparable to return prediction
- transfer learning
  from statistical prediction to economically motivated trading tasks

Economically important frontier in portfolio research

Banerjee, A., X. Guo, and H. Wang. 2005. On the optimality of conditional expectation as a Bregman predictor. IEEE Transactions on Information Theory 51:2664–9. Conference Name: IEEE Transactions on Information Theory.

Frazzini, A., R. Israel, and T. J. Moskowitz. 2018. Trading Costs. doi:10.2139/ssrn.3229719.

Kyle, A. S. 1985. Continuous auctions and insider trading. Econometrica 53:1315–35. Publisher: [Wiley, Econometric Society].

Patton, A. J. 2020. Comparing Possibly Misspecified Forecasts. Journal of Business & Economic Statistics 38:796–809. Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/07350015.2019.1585256.

# Trading Volume Alpha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| *McGill* | *Yale* | *Yale* | *JHU* | HKUST (GZ) |


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# APPENDIX

# Prediction results in size groups and "mixture of experts"

| size group | jointly | nano | micro | small | large | mega |
|---|---|---|---|---|---|---|
| training obs | 2,522,619 | 300,790 | 797,880 | 680,209 | 479,839 | 263,901 |
| testing obs | 1,893,067 | 273,792 | 467,413 | 552,503 | 384,819 | 214,540 |
| A: pooled training evaluated in size groups and jointly | | | | | | |
| ols$_{all}$ | 15.99 | 13.32 | 12.60 | 20.90 | 25.49 | 26.16 |
| nn$_{all}$ | 18.45 | 15.80 | 14.86 | 23.71 | 27.76 | 29.12 |
| rnn$_{all}$ | 19.86 | 16.63 | 16.14 | 26.00 | 30.50 | 32.02 |
| B: size group training evaluated in size groups and jointly (mixture of experts) | | | | | | |
| ols+moe$_{all}$ | 16.34 | 13.68 | 12.73 | 21.43 | 25.93 | 27.47 |
| nn+moe$_{all}$ | 17.78 | 15.29 | 14.43 | 22.69 | 26.57 | 27.71 |
| rnn+moe$_{all}$ | 18.26 | 15.24 | 14.71 | 24.76 | 29.02 | 30.99 |

# Portfolio strategy based on volume prediction, overview

Portfolio problem in two steps

1. given targets positions $\left\{x^*_{i,t}\right\}$
   - informed by return signals, not our focus, taken as given
   - solution is general to any target, will evaluate several empirically

2. choose optimal trading rate based on volume forecasts
   - how aggressively to track the target
   - balancing the trade-off between tracking error and trading cost

# Microfoundation of trading model

Quadratic tracking error as the result of mean-variance optimization

$$A(1+r_t^{\mathrm{f}}) + \sum_i x_{i,t} m_{i,t} - \frac{\gamma}{2A} \sum_{i,j} x_{i,t} x_{j,t} \sigma_{ij,t}^2 - \sum_i TradingCost_{i,t},$$

$\Rightarrow$

$$-\frac{\gamma\sigma^2}{2A} \sum_i \left( x_{i,t} - \underbrace{\frac{A}{\gamma\sigma^2} m_{i,t}}_{x_{i,t}^*} \right)^2 - \sum_i TradingCost_{i,t} + \left( A(1+r_t^{\mathrm{f}}) + \frac{A}{2\gamma\sigma^2} \sum_i m_{i,t}^2 \right)$$

(assuming homoscedsticity ($\mathbb{V}\mathrm{ar}_t r_{i,t+1} = \sigma^2$) and zero covariances) trackingerror